

AMMAN CHOUGLE

Mechatronics Engineer | IOT, Design & Automation | Mumbai, India
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PROFESSIONAL SUMMARY

Embedded and Mechatronics Engineer with 3 years of progressive industry experience leading automation and IoT initiatives across industrial and process environments. Currently heading automation and IoT development at Indra, delivering smart control systems, microcontroller-based solutions, motor-driven automation, and real-time data acquisition platforms for enterprise clients including Taj Hotels, HCL, and Tata Group. Adept at replacing traditional PLC architectures with embedded controllers that deliver measurable gains in system responsiveness, diagnostics, and energy efficiency. Holds an MSc in Mechatronics and Robotics from the University of Leeds, UK, complemented by deep industry expertise in telemetry, SCADA, and Modbus TCP/IP networks. Committed to building intelligent, energy-efficient systems that solve real industrial problems at scale.

CORE COMPETENCIES

Electrical & Controls: Electrical Systems Design · PLC Programming (IEC 61131-3) · SCADA/HMI · Control Panel Design · Modbus TCP/IP
Power & Energy: Power Systems Awareness (T&D, Substations) · Energy Monitoring · Net Zero & Sustainability Engineering
IoT & Instrumentation: Industrial IoT Architecture (RS485, Analog input/output, etc) · Sensor Integration (flow, pressure, COD) · PCB Design · Edge-to-Cloud Connectivity (MQTT, HTTP)
Analysis & Tools: Data Analysis · Root Cause Analysis · Feasibility Studies · MATLAB · Python · AutoCAD · SolidWorks · Fusion 360

PROFESSIONAL EXPERIENCE

Mechatronics Engineer — IoT & Product Development · Indra (Inphlox Water Systems Pvt. Ltd.) · Mumbai, India Dec 2024 – Present

- **Delivered full-cycle IoT solutions** for Taj Hotels, HCL, and Tata Group — encompassing requirements scoping, electrical design, sensor integration, on-site commissioning, and formal client handover.
- Engineered complete control system architectures integrating PLCs, multi-parameter sensors (flow, pressure, COD), and Modbus TCP/IP networks; designed and deployed a custom IoT-PLC interface PCB from schematic to live plant environment.
- Conducted systematic operational data analysis to identify process inefficiencies, translating insights into engineering recommendations that measurably improved system performance and reduced energy waste.
- Produced detailed electrical schematics and technical documentation; led commissioning, validation, and acceptance testing to ensure compliance with operational safety and performance standards.
- Coordinated cross-functional teams and third-party vendors to maintain project timelines and quality standards across concurrent deployments.

Automation & Controls Engineer · B&R Industrial Automation · L&T Electrical & Automation · Festo Didactic Nov 2019 – May 2022

- Designed, configured, and maintained industrial electrical and automation systems — PLCs, sensors, HMIs, and distributed control architectures — within fast-paced manufacturing environments.
- Applied structured root cause analysis and fault-finding methodologies to reduce unplanned downtime and improve overall equipment effectiveness (OEE).
- Developed and standardised technical documentation — wiring diagrams, functional specifications, and test protocols — ensuring consistency and auditability across multi-site installations.

EDUCATION

MSc (Engineering) — Mechatronics & Robotics	University of Leeds, UK
Bachelor of Engineering — Mechatronics Engineering	University of Mumbai, India

SELECTED PROJECT HIGHLIGHTS

- **Enterprise IoT Monitoring Platform** — Architected and deployed a multi-site data acquisition and remote monitoring solution, integrating real-time sensor feeds with cloud-based analytics for continuous process visibility and predictive maintenance.
- **Custom IoT-PLC Interface PCB** — Designed a bespoke interface board from schematic to field deployment, enabling reliable Modbus TCP/IP communication between edge IoT devices and legacy industrial PLCs.
- **Box Lift Conveyor System** — Delivered full mechanical design, motor selection, and PLC sequencing logic for an automated material-handling system, achieving safe and repeatable production-line operation.
- **Continuum Robotic Arm (MSc)** — Modular 3D-printed arm with embedded control electronics — integrating kinematics, additive manufacturing, and embedded systems design.